Appendix A

Major Project Proposal (Individual Mode)

Title: Email Spam Detection Using Machine Learning

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Name, Designation and Communication details of the Guide: Project Guide: ____________________________

Abstract

Email Spam Detection is an intelligent text classification system designed to automatically identify unwanted or fraudulent emails and separate them from legitimate emails. The proposed project applies Machine Learning and Natural Language Processing techniques to analyze email content, subject lines, and metadata. Algorithms such as Naive Bayes, Logistic Regression, Support Vector Machine, and Random Forest are trained on labeled datasets to classify emails as Spam or Ham. The system improves email security, reduces manual filtering effort, and helps users avoid phishing attacks and malicious content. This project demonstrates the practical use of data science in real-world communication systems.

Assumptions / Declarations

1. Publicly available email datasets will be used for academic purposes only.
2. Model accuracy depends on quality of training data and preprocessing methods.
3. The project will be developed using Python and open-source libraries.
4. The system is intended for educational demonstration and research.

Main Objective / Deliverable

• To collect and preprocess spam/ham email datasets.
• To extract features using TF-IDF, Bag of Words, and text statistics.
• To train and compare multiple machine learning classification models.
• To achieve high Accuracy, Precision, Recall, and F1-Score.
• To build a user-friendly interface for real-time email prediction.
• To generate reports and visualization of model performance.

Timeline and Milestones

Milestones	Timeline
Problem Definition and Requirement Analysis	Week 1
Dataset Collection and Data Cleaning	Week 2-3
Feature Extraction and Text Preprocessing	Week 4-5
Model Building and Training	Week 6-8
Testing, Evaluation and Optimization	Week 9-10
GUI Development and Integration	Week 11
Documentation and Final Submission	Week 12

Tools to be used

• Software: Python 3.x, Jupyter Notebook / VS Code
• Libraries: Pandas, NumPy, Scikit-learn, NLTK, Matplotlib, Seaborn
• Database/File Support: CSV / Excel datasets
• Hardware: Laptop/Desktop with minimum 4GB RAM and Internet Access

Learning involved

- Text classification basics
- Natural Language Processing techniques
- Naive Bayes and supervised learning models
- Feature extraction from text data
- Performance metrics and model evaluation
- Real-time deployment concepts

Date: ________________

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